

GABRIEL ASENCIOS

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OBJECTIVE

Computer Science student with main interests in machine learning and data science, with hands-on experience building predictive models, data pipelines, and deep learning systems. Seeking to apply analytical and ML expertise in research, applied, or data-driven roles.

EDUCATION

Bachelor of Computer Science, Concordia University Expected 2027

Relevant Courses: Data Structures and Algorithms, Machine Learning, Deep Learning, Big Data Analytics, Web Programming, Computer Networks, Databases, and UI/UX.

PROJECTS

FocusGuard – Multimodal Distraction Detection. [\[LINK\]](#)

Core: Python | PyTorch | YOLOv8 | ResNet18

- Developed a hybrid computer vision pipeline that monitors user focus via synchronized **YOLOv8** webcam detection and a **ResNet18** screen classifier trained on a custom dataset collected manually.
- Optimized the **ResNet18** classifier using a hyperparameter sweep, identifying **AdamW** with a **0.001 learning rate** and **0.5 dropout** as the ideal configuration.
- Implemented distraction event handling with configurable confidence thresholds, grace periods, and persistence timers to suppress false positives and ensure reliable real-time alerts.

Retrieval Augmented Time Series Forecasting. [\[LINK\]](#)

Core: Pytorch | Python

- Replicated the RAFT framework, a RAG-inspired time series forecaster combining multi-scale retrieval with a lightweight MLP, achieving MSE results within **0.02** of the original paper across **10** benchmark datasets.
- Extended evaluation to a real-world e-commerce sales dataset, establishing a new benchmark on highly non-stationary business data (MSE: **1.904**, MAE: **1.085**), demonstrating model generalization beyond standard datasets.

Customer Churn Prediction – Decision Trees & Random Forests. [\[LINK\]](#)

Core: Python | Pytorch | Pandas

- Preprocessed and explored a customer churn dataset through EDA, duplicate removal, one-hot encoding, and stratified splitting, then trained and tuned Decision Tree and Random Forest classifiers using GridSearchCV to optimize F1-score.
- Conducted feature importance analysis identifying watch time, resolution time, and complaint volume as top churn predictors, translating model outputs into three concrete business retention recommendations.

SmartCart – E-Commerce Recommender System. [\[LINK\]](#)

Core: Python | Pytorch | Pandas

- Cleaned and merged two e-commerce datasets, built a user-item matrix, and implemented a user-based collaborative filtering system using cosine similarity, evaluated at Precision@5: **0.032** and Recall@5: **0.05**.
- Applied the Apriori algorithm for association rule mining, uncovering cross-category purchase patterns and product bundling opportunities using support, confidence, and lift metrics.

HACKATHONS

GymBud — ConUHacks X [\[LINK\]](#)

Core: *React* | *Node.js* | *Express* | *MongoDB*

- Developed a personalized AI fitness coach using the Gemini API, React, and MongoDB to generate custom gym routines and deliver context-aware guidance through a responsive Tailwind CSS interface.
- Built and presented the project to a panel of industry judges within 24 hours.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks	React, Node.js, Express, Tailwind CSS
Libraries	PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Tools/Cloud	REST APIs, MongoDB, MySQL, Docker, Git, GitHub, Google Colab, AWS

WORK EXPERIENCE

Bartender/Runner
LOV McGill

Sept 2022 - April 2024
Montreal, QC

- Collaborated with cross-functional teams in a fast-paced environment, strengthening multitasking, problem-solving, and communication skills under pressure.